

★ Estimation

Since Y is binary and rare

$$\log(E(Y|A, M, C)) = \theta_0 + \theta_1 A + \theta_2 M + \theta_c C$$

$$Q(a, a^*) = \int_c \int_m E(Y|M=m, A=a, C=c) P(M=m|A=a^*, C=c) P(C=c)$$

$$= \int_c \int_m \left\{ e^{\theta_0 + \theta_1 a + \theta_2 m + \theta_c c} \right\} P(M=m|A=a^*, C=c) P(C=c)$$

$$= e^{\theta_0 + \theta_1 a} \int_c e^{\theta_c c} E(e^{\theta_2 M} | A=a^*, C=c) P(C=c)$$

If M is conti, $M \sim N(\mu_M, \sigma^2)$

$$\mu_M = E(M|A=a^*, C=c) = \alpha_0 + \alpha_1 a^* + \alpha_c c$$

$$\begin{aligned} \text{then } E(e^{\theta_2 M} | A=a^*, C=c) &= e^{\mu_M \theta_2 + \frac{1}{2} \sigma^2 \theta_2^2} \\ &= e^{\theta_2 (\alpha_0 + \alpha_1 a^* + \alpha_c c) + \frac{1}{2} \sigma^2 \theta_2^2} \end{aligned}$$

$$\begin{aligned} \Rightarrow Q(a, a^*) &= e^{\theta_0 + \theta_1 a} E \left[e^{\theta_2 (\alpha_0 + \alpha_1 a^* + \alpha_c c) + \frac{1}{2} \sigma^2 \theta_2^2} \right] \\ &= e^{\theta_0 + \theta_1 a + \theta_2 (\alpha_0 + \alpha_1 a^*) + \frac{1}{2} \sigma^2 \theta_2^2} E(e^{(\alpha_c \theta_2 + \theta_c) c}) \end{aligned}$$

$$NDE_{RR} = \frac{e^{\theta_0 + \theta_1 + \alpha_0 \theta_2 + \frac{1}{2} \sigma^2 \theta_2^2}}{e^{\theta_0 + \alpha_0 \theta_2 + \frac{1}{2} \sigma^2 \theta_2^2}} = e^{\theta_1}$$

$$NIE_{RR} = e^{\alpha_1 \theta_2}$$

If M is Binary $\logit[P(M=1|A=a^*, C=c)] = \alpha_0 + \alpha_1 a^* + \alpha_c c$

$$\begin{aligned} \text{define } P(M=1|A=a^*, C=c) &= \text{logit}^{-1}(\alpha_0 + \alpha_1 a^* + \alpha_c c) \\ &= \pi(a^*, c, \alpha) \end{aligned}$$

$$P(M=0|A=a^*, C=c) = 1 - \pi(a^*, c, \alpha)$$

$$\begin{aligned} \text{then } E(e^{\theta_2 M} | A=a^*, C=c) &= e^{\theta_2} \pi(a^*, c, \alpha) + e^0 [1 - \pi(a^*, c, \alpha)] \\ &= 1 + (e^{\theta_2} - 1) \pi(a^*, c, \alpha) \end{aligned}$$

- If outcome is rare and M follows a linear regression, the difference method is approximately equal to the $\log(NIE_{RR})$

Let T be time-to-event outcome.

(Measure the amount time from start pt until a specific event occurs)

A, M, C remain unchanged.

* Recap.

$$T \sim f(t)$$

$S(t) = P(T > t) = 1 - F(t)$ is the survival fn at time t .



$$S(t|c) = P(T > t | c)$$

$$-\frac{dS(t)}{dt} = f(t)$$

$$\lambda(t) = \lim_{\Delta t \rightarrow 0^+} \frac{P(t \leq T < t + \Delta t | T \geq t)}{\Delta t} = \frac{f(t)}{S(t)} = -\frac{d \ln S(t)}{dt}$$

$$\Rightarrow e^{-\Lambda(t)} = e^{-\int_0^t \lambda(s) ds} = S(t)$$

hazard at time t

{ Instantaneous rate of the event conditional on $T \geq t$

* Counterfactuals

$$\begin{cases} T(a) \\ M(a) \\ T(a, m) \end{cases} \text{ is counterfactual } \begin{cases} \text{event time} \\ \text{mediator} \\ \text{event time} \end{cases} \text{ if } \begin{cases} A=a \\ A=a \\ A=a, M=m \end{cases}$$

$T(a, M(a^*))$ is nested counterfactual event time if $A=a$ and mediator had been set the level it would had been $A=a^*$

* There are multiple ways or scales by which we might decompose a total effect comparing exposure levels a and a^* into direct and indirect effect.

Ex. log hazard.

$$\log \{ \lambda_{T(a)}(t) \} - \log \{ \lambda_{T(a^*)}(t) \} = \left[\log \{ \lambda_{T(a, M(a))}(t) \} - \log \{ \lambda_{T(a, M(a^*))}(t) \} \right] \\ + \left[\log \{ \lambda_{T(a, M(a^*))}(t) \} - \log \{ \lambda_{T(a^*, M(a^*))}(t) \} \right]$$

$$\Downarrow$$

$$\frac{\lambda_{T(a)}(t)}{\lambda_{T(a^*)}(t)} = \frac{\lambda_{T(a, M(a))}(t)}{\lambda_{T(a, M(a^*))}(t)} \times \frac{\lambda_{T(a, M(a^*))}(t)}{\lambda_{T(a^*, M(a^*))}(t)}$$

We have that

$$\lambda_{T(a, M(a^*))}(t) = \frac{f_{T(a, M(a^*))}(t)}{S_{T(a, M(a^*))}(t)}$$

$$f_{T(a, M(a^*))}(t) = \int f_{T(a, M(a^*))}(t | M(a^*)=m) dF_{M(a^*)}(m) \\ = \int f_{T(a, m)}(t | M(a^*)=m) dF_{M(a^*)}(m) \\ = \int f_{T(a, m)}(t | A=a, M=m) dF_{M(a^*)}(m) \\ = \int f_T(t | a, m) dF_{M(a^*)}(m)$$

$$S_{T(a, M(a^*))}(t) = \int S_T(t | a, m) dF_{M(a^*)}(m) = \int e^{-\Lambda(t | a, m)} dF_{M(a^*)}(m)$$

Consider Proportional Hazard Model

$$\lambda_T(t | a, m) = \underbrace{\lambda_T(t | 0, 0)}_{\text{underlying hazard}} e^{\theta_1 a + \theta_2 m}$$

and mediator model

$$E(M | A) = \beta_0 + \beta_1 A$$

then

$$f_{T(a, M(a^*))}(t) = \int \lambda_T(t | 0, 0) e^{\theta_1 a + \theta_2 m} e^{-[\Lambda(t | 0, 0) e^{\theta_1 a + \theta_2 m}]} dF_{M(a^*)}(m)$$

$$S_{T(a, M(a^*))}(t) = \int e^{-[\Lambda(t | 0, 0) e^{\theta_1 a + \theta_2 m}]} dF_{M(a^*)}(m)$$

$$\Rightarrow \lambda_{T(a, M(a^*))} \approx \lambda_T(t | 0, 0) e^{\theta_1 a + \theta_2 (\beta_1 + \beta_2 a^*) + \frac{1}{2} \theta_2^2 \Delta^2} \\ = \left(\lambda_T(t | 0, 0) e^{\theta_2 \beta_1 + \frac{1}{2} \theta_2^2 \Delta^2} \right) e^{\theta_1 a + \theta_2 \beta_2 a^*}$$

$$\frac{\lambda_{T(1, M(1))}(t)}{\lambda_{T(1, M(0))}(t)} = e^{\theta_2 \beta_2}$$

$$\frac{\lambda_{T(1, M(0))}(t)}{\lambda_{T(0, M(0))}(t)} = e^{\theta_1}$$

★ Difference method is lower bdd for NIEOR (Underestimate)

$$\begin{pmatrix} \text{logit}(p) = x \\ \Downarrow e^x \\ p = \frac{e^x}{1+e^x} \end{pmatrix}$$

$$\text{logit}^{-1}(\theta_0 + \theta_1 a + \theta_2 m + \theta_3 c) = \frac{e^{\theta_0 + \theta_1 a + \theta_2 m + \theta_3 c}}{1 + e^{\theta_0 + \theta_1 a + \theta_2 m + \theta_3 c}}$$

$$P(Y=1 | M=m, A=a, C=c)$$

$$E[Y(a, M(a^*)) | C] = \int_m E(Y | M=m, A=a, C=c) f_{M|A,C}(m | a^*, c) dm$$

$$P(Y(a, M(a^*))=1 | C) = \phi(a, a^*)$$

$$NDE_{OR} = \left[\frac{\phi(a, a^*)}{1 - \phi(a, a^*)} \right] / \left[\frac{\phi(a^*, a^*)}{1 - \phi(a^*, a^*)} \right]$$

$$\text{Let } A = 1 - \phi(a, a^*) = \int_m \frac{1}{1 + e^{\theta_0 + \theta_1 a + \theta_2 m + \theta_3 c}} f_{M|A,C}(m | a^*, c) dm$$

$$= E_{M|A^*, C} \left(\frac{1}{1 + e^{\theta_0 + \theta_1 a + \theta_2 m + \theta_3 c}} \right)$$

Similar

$$B = E_{M|A^*, C} \left(\frac{1}{1 + e^{\theta_0 + \theta_1 a^* + \theta_2 m + \theta_3 c}} \right)$$

$$NDE_{OR} = \left(\frac{1}{A} - 1 \right) / \left(\frac{1}{B} - 1 \right) \quad \left(NDE_{OR} \leq e^{\theta_1(a-a^*)} \right. \\ \left. e^{\theta_1(a-a^*)} \left(\frac{1}{B} - 1 \right) - \left(\frac{1}{A} - 1 \right) \geq 0 \right)$$

Consider

$$e^{\theta_1(a-a^*)} \left(\frac{1}{B} - 1 \right) - \left(\frac{1}{A} - 1 \right) \\ = \frac{A e^{\theta_1(a-a^*)} - B}{AB} - (e^{\theta_1(a-a^*)} - 1) = (\Delta)$$

$$A e^{\theta_1(a-a^*)} - B = E_{M|A^*, C} \left(\frac{e^{\theta_1(a-a^*)} [1 + e^{\theta_0 + \theta_1 a + \theta_2 m + \theta_3 c}] - [1 + e^{\theta_0 + \theta_1 a^* + \theta_2 m + \theta_3 c}]}{(1 + e^{\theta_0 + \theta_1 a + \theta_2 m + \theta_3 c})(1 + e^{\theta_0 + \theta_1 a^* + \theta_2 m + \theta_3 c})} \right)$$

$$= [e^{\theta_1(a-a^*)} - 1] C$$

$$(\Delta) = \left(\frac{C}{AB} - 1 \right) (e^{\theta_1(a-a^*)} - 1)$$

Lemma 1. Let f and g both non-decreasing

$$\text{Cov}(f(x), g(x)) \geq 0$$

$$C - AB \geq 0$$

If $\theta_1 > 0$, we have

$$\left(\frac{C}{AB} - 1\right)(e^{\theta_1(a-a^*)} - 1) \geq 0$$

$$\begin{aligned} &\Downarrow \\ e^{\theta_1(a-a^*)} &\geq NDE_{OR} \end{aligned}$$

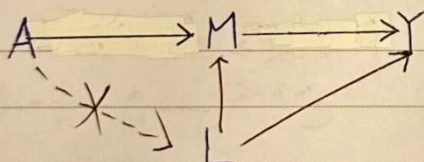
From the property that

$$TE_{OR} = NIE_{OR} \times NDE_{OR}$$

$$\frac{e^{g(a-a^*)}}{NDE_{OR}} = NIE_{OR} \geq e^{(g_1 - \theta_1)(a-a^*)}$$

$$\left(\begin{array}{l} \text{Remark. } CDE_{OR} = e^{\theta_1(a-a^*)} \geq NDE_{OR} \\ PE_{OR} = e^{(g_1 - \theta_1)(a-a^*)} \leq NIE_{OR} \end{array} \right)$$

Interventional Approach



$$\Leftrightarrow Y(a, m) \perp\!\!\!\perp M(a^*) | C$$

- The cross would exchangeability is fail when exposure affect M-Y confounder.

★ Let G denote random draws from $\text{dist}^n M$
 $G(a)$ is random draws of $M(a)$

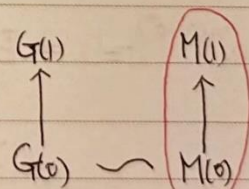
1) $G \perp\!\!\!\perp A$

2) $F_G(u) = F_M(u)$, $G \sim M$

Alternative definitions for NDE and NIE is randomly interventional analogues of NDE and NIE

$$IIE = E[Y(a, G(a))] - E[Y(a, G(a^*))]$$

$$IDE = E[Y(a, G(a^*))] - E[Y(a^*, G(a^*))]$$



Same individual

$$\psi(a, a^*) = E[Y(a, G(a^*))]$$

$$IIE = \psi(a, a) - \psi(a, a^*)$$

$$IDE = \psi(a, a^*) - \psi(a^*, a^*)$$

★ Identification of $\psi(a, a^*)$

$$\psi(a, a^*) = E[Y(a, G(a^*))]$$

$$= \int_C E[Y(a, G(a^*)) | C=c] P(C=c)$$

$$= \int_C \left\{ \int_m E[Y(a, G(a^*)) | G(a^*)=m, C=c] P(G(a^*)=m | C=c) \right\} P(C=c)$$

$$(Consistency) = \int_{c,m} E[Y(a, m) | G(a^*)=m, C=c] P(G(a^*)=m | C=c) P(C=c)$$

$$(G(a^*) \perp\!\!\!\perp Y(a, m) | C) = \int_{c,m} E[Y(a, m) | C=c] P(G(a^*)=m | C=c) P(C=c)$$

$$(G(a^*) \sim M(a^*) | C) = \int_{c,m} E[Y(a, m) | C=c] P(M(a^*)=m | C=c) P(C=c)$$

$$(M(a^*) \perp\!\!\!\perp A | C) = \int_{c,m} E[Y(a, m) | C=c] P(M(a^*)=m | A=a^*, C=c) P(C=c)$$

$$(Consistency) = \int_{c,m} E[Y(a, m) | C=c] P(M=m | A=a^*, C=c) P(C=c)$$

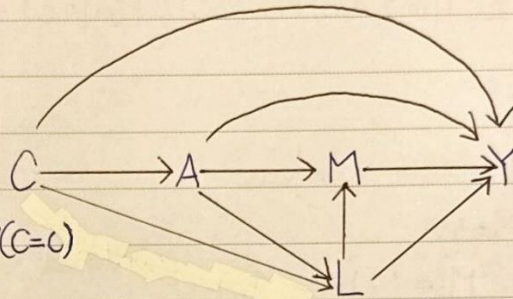
$$(Y(a, m) \perp\!\!\!\perp A | C) = \int_{c,m} E[Y(a, m) | A=a, C=c] P(M=m | A=a^*, C=c) P(C=c)$$

$$= \int_{c,m} \left\{ \int_l E[Y(a, m) | L=l, A=a, C=c] P(L=l | A=a, C=c) \right\} \times P(M=m | A=a^*, C=c) P(C=c)$$

$$(Y(a, m) \perp\!\!\!\perp M | A, L, C)$$

$$= \int_{c,m,l} E[Y(a, m) | M=m, L=l, A=a, C=c] P(L=l | A=a, C=c)$$

$$(Consistency) = \int_{c,m,l} E[Y(a, m) | M=m, L=l, A=a, C=c] P(L=l | A=a, C=c) \times P(M=m | A=a^*, C=c) P(C=c)$$



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$$IDE = \int_{c,m,l} \left\{ E(Y|m,l,a,c) P(l|a,c) - E(Y|m,l,a^*,c) P(l|a^*,c) \right\} \times P(m|a^*,c) P(c)$$

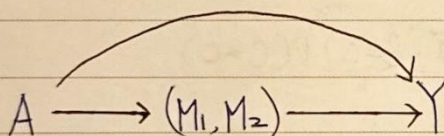
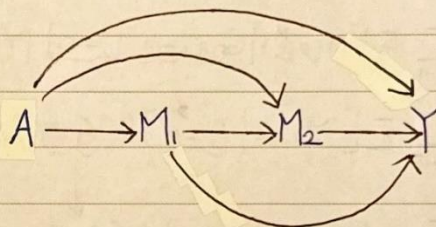
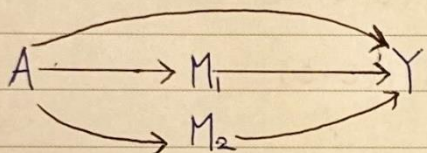
$$IIE = \int_{c,m,l} E(Y|m,l,a,c) P(l|a,c) \left\{ P(m|a,c) - P(m|a^*,c) \right\} P(c)$$

Multiple Mediation Analysis

Multiple Mediator

$$M = (M_1, \dots, M_p) \begin{cases} \text{Ordered} \\ \text{Non-ordered} \end{cases}$$

DAGs



★ Path Specific Effect (PSE)

Evaluate the effect mediated by a certain combination of Mediators.

In general, k mediators $\Rightarrow 2^k$ PSEs. (increase exponentially)

Ex. $A \rightarrow Y$ (DE)	(0,0)	path not mediated by M_1 or M_2
$A \rightarrow M_1 \rightarrow Y$	(1,0)	path mediated by M_1 only
$A \rightarrow M_2 \rightarrow Y$	(0,1)	" M_2 "
$A \rightarrow M_1 \rightarrow M_2 \rightarrow Y$	(1,1)	path mediated by M_1 then M_2

$$I = (I(M_1), I(M_2))$$

decompose TE into $\begin{cases} 2^{k-1} \text{ indirect effect} \\ 1 \text{ direct effect} \end{cases}$